KINGSON EZERUIGBO

MSC ARTIFICIAL INTELLIGENCE STUDENT

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https://www.github.com/NedytheCoder

PROFILE

- MSc Artificial Intelligence student at the National College of Ireland (NCI) with hands-on experience in data exploration, preprocessing, and model development across multiple academic labs and applied projects. Experienced in training, validating, and evaluating machine learning models, with a solid understanding of end-to-end ML workflows including feature engineering, model selection, performance tuning, and documentation of use cases. Skilled in designing and implementing supervised and unsupervised learning models, data analysis, and experimental design.
- Strong foundation in linear algebra, probability, and statistics, supported by a background in Computer Science and Software Engineering. Three years of web development experience have strengthened my proficiency in Python, SQL, Git, and collaborative development workflows, enabling me to contribute effectively to data-driven and AI-focused initiatives. Fluent in English and French, with communication skills in cross-functional and international environments.

TECHNOLOGY SKILLS

- **Python (Data Analysis & Machine Learning):** Data manipulation and preprocessing using NumPy and Pandas, Exploratory Data Analysis (EDA), Data visualization (Matplotlib / Seaborn), Implementation of supervised learning models (e.g., regression and classification) using scikit-learn, Model training and evaluation (train/test split, cross-validation, accuracy, precision, recall), Hyperparameter optimization using GridSearchCV / RandomizedSearchCV, Feature engineering fundamentals, Model comparison and performance improvement, Working with structured datasets (CSV/JSON), Writing modular and reusable code.
- **Next.Js (AI-Ready Frontend Systems):** Development of dynamic web applications using SSR and SSG, Component-based architecture, Dynamic routing, State management, API integration (REST), Handling asynchronous data pipelines, Rendering and visualizing structured data, Performance optimization (code splitting, lazy loading), SEO optimization.
- **Data Handling & API Integration:** Consuming and transforming JSON data, managing asynchronous data flows, Error handling in data-driven interfaces, Designing scalable UI layers for dynamic backend/model outputs.
- **JavaScript / TypeScript (Software Engineering Foundations):** ES6+ features, Async/await, Modular architecture, Type safety, clean code practices, Debugging and performance profiling, Building maintainable large-scale frontend codebases.
- **Performance & Optimization:** Reducing render cycles, Bundle size optimization, Lighthouse performance auditing, Image optimization, improving client-side execution efficiency which are transferable to optimizing AI-powered web applications.
- **Collaboration & Engineering Practices:** Git, branching strategies, Code reviews, Agile development workflow, Writing readable and maintainable production-level code.
- **Certifications:**
  - Project Management Professional (PMP), Project Management Institute 2025
  - Google Data Analytics 2025
  - Advanced Diploma in Software Engineering (ADSE) 2023
- **Additional Certifications:**
  - Diploma in French Language studies B2 (DELF B2) 2025

WORK EXPERIENCE

11/2024 - 01/2026 | Nelson Benjamin LTD **Software Engineer**

- Developed data-driven web applications using Next.js and React, building dynamic interfaces capable of rendering structured and unstructured API responses.
- Optimized client-side performance through code splitting and lazy loading, improving rendering efficiency for data-heavy interfaces.
- Integrated REST APIs and managed asynchronous data flows using async/await, ensuring reliable handling of dynamic backend outputs.
- Implemented server-side rendering (SSR) to improve performance and SEO, demonstrating understanding of production-level web architecture.
- Designed reusable UI systems to support modular expansion which are transferable to AI model dashboards and analytics platforms.

- Debugged and profiled front-end performance bottlenecks, improving responsiveness and user experience under varying data loads.
- Collaborated using Git-based workflows to maintain clean, maintainable code in iterative development environments.

11/2023 - 11/2024 | Nelson Benjamin LTD **Software Engineer Intern**

- Developed and maintained production-level web applications using Next.js and React within a collaborative engineering team environment.
- Implemented dynamic routing, reusable component architecture, and server-side rendering (SSR) to improve scalability and performance.
- Integrated REST APIs and managed asynchronous data flows, handling edge cases such as loading states, error handling, and data consistency.
- Collaborated with backend developers and designers to translate business requirements into functional, user-friendly interfaces.
- Optimized frontend performance through code splitting, lazy loading, and bundle size reduction techniques.
- Participated in code reviews, version control workflows (Git), and Agile sprint cycles to ensure high-quality, maintainable code.
- Debugged and resolved production issues using browser developer tools, improving application stability and user experience.

P R O J E C T S

Transformer-Based Early Nadir Frequency Prediction https://github.com/NedytheCoder/transformer-early-nadir-frequency-prediction

- Built an encoder-only transformer (4 layers, 8 attention heads, 2.17M parameters) to predict post-disturbance frequency nadir depth and timing from short PMU windows, trained on 38,412 ANDES simulations of N-1 generator trips on the IEEE 39-bus low-inertia system.
- Benchmarked the model against LSTM, CNN, SPM-BPNN, ridge, and an analytical SFR baseline across 56 seeded runs and four observation windows, reaching 38.4 mHz MAE and R^2 0.930 on nadir depth at 500 ms.
- Identified graceful degradation under measurement noise as the transformer's decisive advantage (175 mHz MAE at $\sigma = 0.01$ Hz against 9,248 mHz for the BPNN baseline), while reporting that windows below 100 ms separate no architecture from a linear floor.

Irish SEM TLAF Clustering & Explainability Study https://github.com/NedytheCoder/irish-sem-unsupervised-analysis

- Proposed an unsupervised ML study of Irish SEM TLAFs by combining EirGrid and SONI TLAF records with geospatial plant data.
- Benchmarked K-Means, DBSCAN, Hierarchical Agglomerative Clustering, and Gaussian Mixture Models using internal validity and bootstrapped stability metrics.
- Designed an XGBoost and SHAP workflow to explain high-loss group membership and translate clusters into infrastructure recommendations.

Material Procurement Cost Optimization Analysis https://github.com/NedytheCoder/material_cost_optimization_analysis

- Built a two-stage machine learning pipeline combining TED public procurement data and materials datasets to support contract cost prediction and lower-cost material substitution analysis.
- Cleaned, translated, transformed, and modelled procurement data using regression models including Random Forest, Gradient Boosting, Ridge, nd XGBoost, evaluted with RMSE, MAE, MAPE, and R^2
- Developed a materials intelligence layer using World Bank commodity-price proxies, confidence scoring, KMeans clustering, DBSCAN comparison, and nearest-neighbour search to identify technically similar lower-cost alternatives.

E D U C A T I O N

Jan 26 - Jan 27 | National College of Ireland **MSc Artificial Intelligence**

Modules: Foundations of Artificial Intelligence, Programming for Artificial Intelligence, Data Analytics for Artificial Intelligence, Data Governance and Ethics, Engineering and Evaluating Artificial Intelligence Systems, Intelligent Agents and Process Automation, Artificial Intelligence Driven Decision Making, Machine Learning, Emergent Artificial Intelligence Technologies and Sustainability

2019 - 2024 | Enugu State University of Science and Technology **BSc (Hons) Computing 2nd Class Upper Division**

2012 - 2017 | Bishop Otubelu Juniorate **WAEC Leaving Certificate**

I N T E R E S T S A N D A C H I E V E M E N T S

French Speaker

- Fluent French Speaker
- Provided language support and mentorship to students preparing for French certification exams.
- Participated in community-based French cultural and language events.

Fitness

- Consistently engaged in strength training and physical fitness (2–3 times per week), demonstrating discipline, goal-setting, and long-term commitment.

Literature

- Regular reader of personal development and behavioral psychology literature, including the 48 Laws of Power by Robert Greene and The Art of Being Lazy.
- Interested in human behavior, decision-making, and personal growth.